

Taiwan Monthly Revenue Surprise Strategy

Portfolio Research Report — Research / Paper-Trading Only

Generated from portfolio_project Markdown documents. This document is not investment advice and does not claim production-ready live trading alpha.

Source directory: /Users/liuynzhen/quant-research/tw_monthly_revenue/portfolio_project

Final Portfolio Index — Taiwan Monthly Revenue Surprise Strategy

Elevator pitch

This project tests whether Taiwan monthly revenue disclosures create post-announcement drift in equities with persistent positive revenue surprise. The research starts from a market-structure hypothesis, builds official-data pipelines, evaluates SUR-style signals, and then subjects the best candidate to walk-forward, sector, winner-dependence, cost, liquidity, timing, and paper-trading gates.

Recommended reading order

1. [README.md](#) (README.md) — project overview.
2. [02_strategy_card.md](#) (02_strategy_card.md) — concise strategy summary.
3. [04_robustness_dashboard.md](#) (04_robustness_dashboard.md) — main evidence and stress tests.
4. [05_execution_and_paper_trading.md](#) (05_execution_and_paper_trading.md) — why it is not production-ready yet.
5. [07_interview_qna.md](#) (07_interview_qna.md) — interview framing.
6. [reproducibility.md](#) (reproducibility.md) — paths and scripts.

Portfolio positioning

Use this project to demonstrate:

- causal hypothesis design: monthly revenue disclosure → delayed repricing;
- Taiwan market data familiarity;
- robust backtest discipline;
- honest rejection / non-promotion of fragile variants;
- execution-aware thinking;
- readiness to move from backtest to paper-trading audit.

Do not overclaim

Recommended public wording:

A portfolio-grade quant research project on Taiwan monthly revenue surprise and short-horizon post-disclosure drift, with an incumbent candidate showing attractive proxy results but still requiring exact announcement timestamps, stronger universe controls, execution simulation, and paper-trading validation before any live-trading claim.

Avoid saying:

- "production-ready alpha";
- "live trading system";
- "guaranteed Sharpe 2+";
- "broad-market Taiwan stock strategy".

Key artifacts in this folder

- README.md
- 01_methodology_report.md
- 02_strategy_card.md
- 03_research_timeline.md
- 04_robustness_dashboard.md
- 05_execution_and_paper_trading.md
- 06_limitations_and_next_steps.md
- 07_interview_qna.md
- 08_paper_trading_template.md
- paper_trading_log_template.csv
- reproducibility.md
- INDEX.md
- FINAL_PORTFOLIO_INDEX.md

Linked source artifacts outside this folder

- ../reports/promising_strategy_registry.md
- ../reports/phase3_12_walkforward_sector_survival_report.md
- ../reports/phase3_18_quiet_digestion_dynamic_sizing_report.md
- ../reports/phase3_19_execution_realism_tradability_report.md
- ../reports/phase3_24_delay_walkforward_oos_sector_stress_report.md
- ../reports/phase3_25_exact_timing_paper_trading_schema_report.md
- ../reports/phase3_26_official_ohlc_limit_parser_report.md
- ../reports/charts/s1_nav_drawdown_zh.png
- ../reports/charts/phase3_12_walkforward_oos_nav_zh.png
- ../reports/charts/phase3_18_dynamic_sizing_nav_zh.png

Final status

- Research portfolio: ready.
- Paper trading: template ready, future live cycles needed.
- Production/live trading: not ready and not claimed.

Taiwan Monthly Revenue Surprise Strategy

Subtitle: From market-structure hypothesis to execution-aware validation for Taiwan equities

Scope: Research / paper-trading project only. No live trading, broker connection, or investment advice.

Executive Summary

This portfolio project studies whether Taiwan equity markets underreact to persistent monthly revenue surprise after public disclosure. Taiwan is unusually suitable for this question because listed and OTC companies disclose monthly revenue, creating a recurring public information event that can be tested systematically.

The final research framing is deliberately conservative:

The strongest candidate is not a broad-market magic formula. It is a Taiwan electronics / semiconductor supply-chain monthly revenue surprise and fundamental momentum research candidate. It is suitable as a portfolio-grade research artifact, but not yet production-ready alpha.

What this project demonstrates

- Building an alpha hypothesis from market structure instead of indicator mining.
- Constructing monthly revenue surprise / SUR-style factors from official data.
- Moving from simple proxy backtests to robustness gates.
- Preserving promising variants instead of overwriting them during search.
- Testing winner dependence, sector dependence, walk-forward OOS, liquidity, costs, and execution realism.
- Knowing when not to promote a strategy despite attractive full-sample results.

Current status

- Portfolio-ready research project: Yes.
- Production-ready trading strategy: No.
- Incumbent candidate: S1 / portfolio-grade v0.1.
- Best final interpretation: Taiwan electronics / semiconductor monthly revenue surprise + fundamental momentum, with quiet-digestion as a research-only sizing diagnostic.

Key headline findings

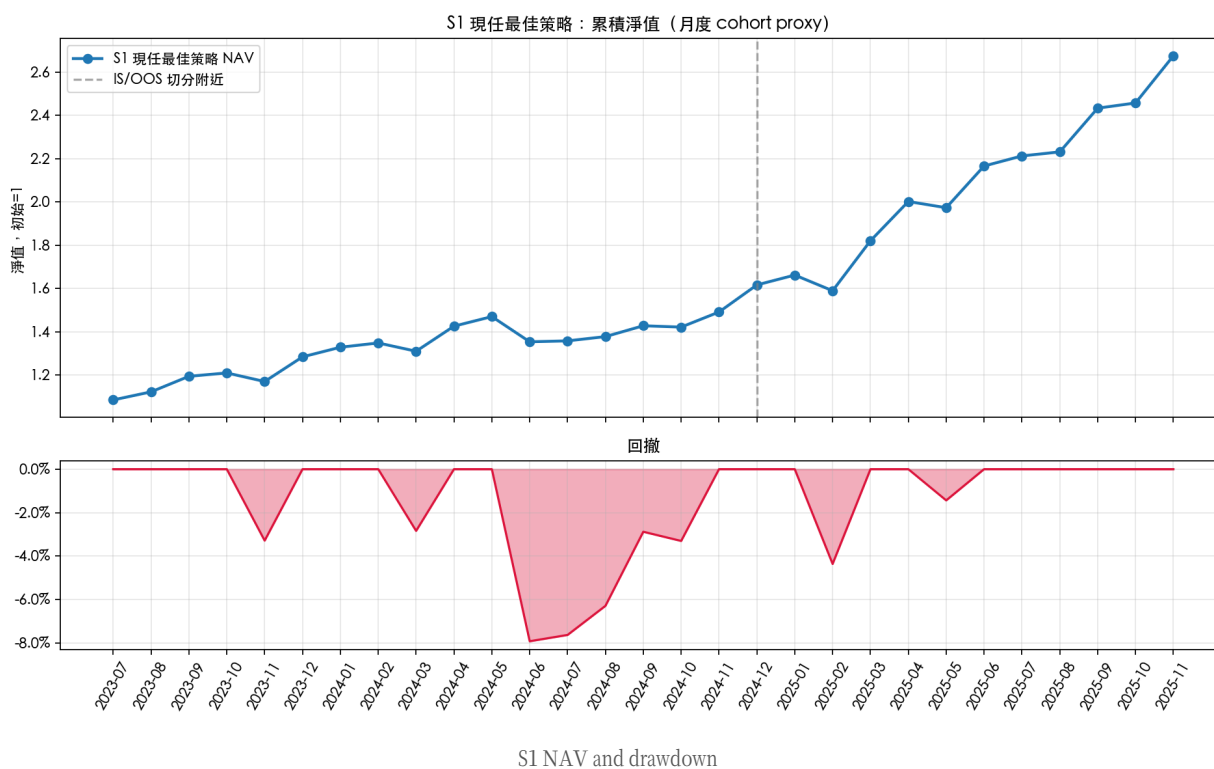
- S1 incumbent proxy result: Sharpe around 2.40, total return around 167.5%, MDD around -7.9% in the earlier portfolio-grade proxy setup.
- Fixed-20 S1 comparator: Sharpe around 1.55, total return around 161.9%, MDD around -21.2%.
- Quiet-digestion dynamic sizing improved the fixed-20 proxy only modestly: Sharpe 1.55 → 1.62, return 161.9% → 174.4%, MDD unchanged around -21.2%.
- Under conservative tradability assumptions (next_open, 1.0% cost, exclude possible limit-up non-fill), quiet boost dropped to return 111.9%, Sharpe 1.24, MDD -24.9%.

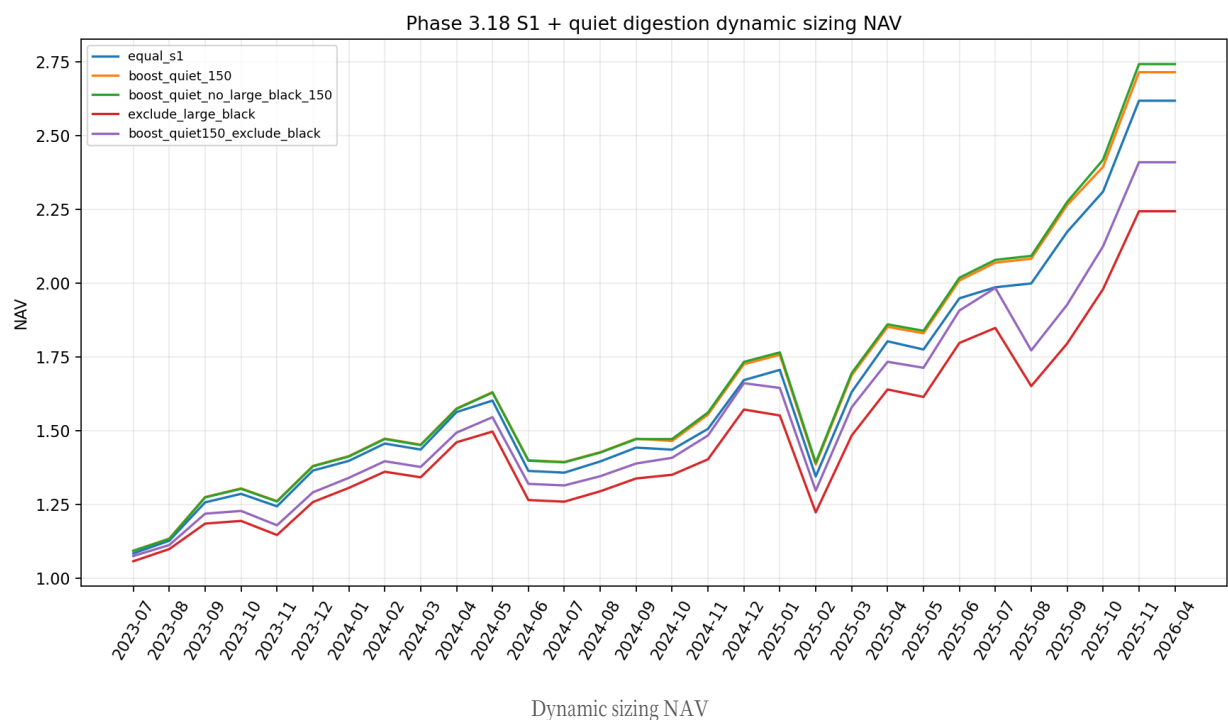
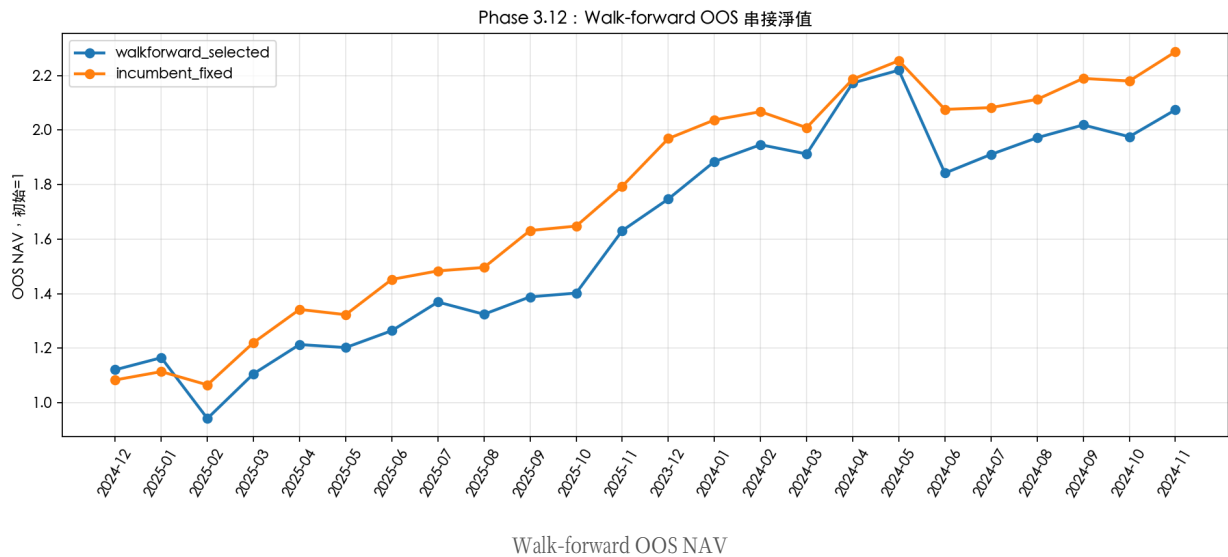
- OOS / sector stress showed strong semiconductor dependence and winner concentration; therefore the strategy is not promoted as broad-market alpha.

Project documents

1. 01_methodology_report.md (01_methodology_report.md) — full project narrative and research process.
2. 02_strategy_card.md (02_strategy_card.md) — one-page strategy card.
3. 03_research_timeline.md (03_research_timeline.md) — phase-by-phase evolution.
4. 04_robustness_dashboard.md (04_robustness_dashboard.md) — key robustness tables and interpretation.
5. 05_execution_and_paper_trading.md (05_execution_and_paper_trading.md) — execution realism and paper-trading plan.
6. 06_limitations_and_next_steps.md (06_limitations_and_next_steps.md) — honest limitations and roadmap.
7. 07_interview_qna.md (07_interview_qna.md) — likely interview challenges and answers.
8. reproducibility.md (reproducibility.md) — scripts, reports, and how to reproduce the research artifacts.

Selected figures





Repository layout

```

portfolio_project/ # Portfolio-ready narrative documents
reports/          # Detailed phase reports and charts
data/processed/   # Processed research outputs / CSV diagnostics
scripts/          # Research scripts for each phase

```

Important disclaimer

This is a research project. It does not place orders, connect to a broker, or recommend live trading. Results are historical/proxy diagnostics and are subject to data timing, survivorship, execution, capacity, and regime risks.

02 — Strategy Card

Strategy name

Taiwan Monthly Revenue Surprise + Fundamental Momentum

Research status

- Status: Portfolio-grade research candidate / not production-ready.
- Incumbent: S1.
- Live trading: No.
- Broker integration: No.

Core hypothesis

Because Taiwanese listed and OTC companies disclose monthly revenue, persistent positive revenue surprise can update expectations about near-term fundamentals. If the market underreacts after disclosure, especially in electronics / semiconductor supply chains, selected stocks may exhibit positive 10 – 20 trading-day drift.

Universe

- Taiwan listed / OTC common-stock-like universe where official revenue and daily market data are available.
- Liquidity screen: typically `avg_turnover_20d >= 50m TWD`; `100m TWD` used as robustness comparator.
- ETFs, warrants, and non-common instruments should be excluded in production-grade research.

Signal

Main S1 family:

high 3M SUR persistence
+ not-overheated 120D-to-20D momentum
+ liquidity screen
+ monthly top-N ranking by SUR core score

Quiet-digestion sizing diagnostic:

low abnormal turnover
+ narrow entry-day range
+ not large-black K

This diagnostic is used only as a research-only sizing hypothesis, not as a standalone promoted strategy.

Portfolio construction

- Monthly rebalance / event-cycle selection after revenue data availability proxy.
- Top 8 names.
- Industry cap = 3.

- Fixed 20 trading-day holding period for main comparator.
- Trailing/stop proxy candidate retained as S1 portfolio-grade v0.1 but still requires execution validation.

Key results

S1 portfolio-grade v0.1 proxy

- Sharpe proxy: about 2.40.
- Total return: about 167.5%.
- MDD: about -7.9%.
- Remove-top-5 Sharpe: about 1.77.

S1 fixed-20 comparator

- Return: 161.9%.
- Sharpe: 1.55.
- MDD: -21.2%.

Quiet-digestion sizing candidate

boost_quiet_no_large_black_150:

- Return: 174.4%.
- Sharpe: 1.62.
- MDD: -21.2%.
- Interpretation: modest sizing improvement, not enough for promotion.

Conservative tradability proxy

next_open + 1.0% cost + exclude possible limit-up risk:

- Equal S1: return 99.8%, Sharpe 1.16, MDD -26.2%.
- Quiet boost: return 111.9%, Sharpe 1.24, MDD -24.9%.

Why it is interesting

- Uses a Taiwan-specific recurring fundamental disclosure.
- Demonstrates a coherent PEAD / underreaction mechanism.
- Shows sector-specific behavior rather than pretending to be universal.
- Includes extensive negative results and failed promotion gates.

Why it is not production-ready

- Current historical revenue table lacks company-level exact announcement timestamps.
- Survivorship / listing-status completeness still needs stronger controls.

- Execution realism still relies on OHLC / limit proxies rather than order-book fills.
- OOS evidence is short: mainly 2023 – 2025.
- Semiconductor and top-winner dependence remain material.
- Paper trading has not yet accumulated multiple cycles of real-time validation.

04 — Robustness Dashboard

Promotion rule

A variant is not promoted just because full-sample Sharpe improves. It must survive:

- year split;
- walk-forward OOS;
- remove-top-winners;
- electronics / semiconductor / no-semiconductor slices;
- liquidity thresholds;
- cost stress;
- execution timing stress.

Core comparison

S1 fixed-20 comparator

- Return: 161.9%.
- Sharpe: 1.55.
- MDD: -21.2%.
- Active months: 29/30.
- Average positions: 7.27.

Quiet boost dynamic sizing

boost_quiet_no_large_black_150:

- Return: 174.4%.
- Sharpe: 1.62.
- MDD: -21.2%.
- Monthly win rate: 76.7%.

Interpretation: modest improvement, but not enough for promotion because MDD did not improve materially.

Remove-winner stress

Equal S1

- Remove 0: Sharpe 1.55.
- Remove 5: Sharpe 1.16.
- Remove 10: Sharpe 0.96.
- Remove 20: Sharpe 0.38.

Quiet boost

- Remove 0: Sharpe 1.62.
- Remove 5: Sharpe 1.22.
- Remove 10: Sharpe 1.04.
- Remove 20: Sharpe 0.48.

Interpretation: quiet boost modestly improves remove-winner profile but does not eliminate top-winner dependence.

Execution realism stress

Conservative proxy:

$\text{next_open} + 1.0\% \text{ all-in cost} + \text{exclude possible limit-up risk}$

Results:

- Equal S1: return 99.8%, Sharpe 1.16, MDD -26.2%.
- Quiet boost: return 111.9%, Sharpe 1.24, MDD -24.9%.

Interpretation: a large portion of headline quality is sensitive to entry timing / execution assumptions.

Walk-forward dynamic sizing

Train 2023 → Test 2024

- Selected rule: `boost_quiet_no_large_black_125`.
- Selected OOS: return 14.45%, Sharpe 0.74.
- Equal S1 OOS: return 12.91%, Sharpe 0.67.

Train 2023 – 2024 → Test 2025

- Selected rule: `boost_quiet_no_large_black_200`.
- Selected OOS: return 40.50%, Sharpe 1.17.
- Equal S1 OOS: return 35.45%, Sharpe 1.08.

Interpretation: OOS improvement exists but is small and based on only two splits.

Sector and winner concentration

2025 selected OOS rule

- All: Sharpe 1.17.
- Semiconductor-only: Sharpe 2.59.
- No-semiconductor: Sharpe 0.31.

- Remove-top-5: Sharpe 0.37.

2024 selected OOS rule

- All: Sharpe 0.76.
- Semiconductor-only: Sharpe 1.52.
- No-semiconductor: Sharpe -0.56.
- Remove-top-5: Sharpe -0.50.

Interpretation: the edge is strongly tied to semiconductor / electronics and a small number of winners. This is the central reason not to call it broad, production-ready alpha.

Capacity proxy

Using monthly portfolio capital capped by participation $\times$ average 20D turnover value:

Equal S1

- ADV 1% median capacity: about 5.2m TWD.
- ADV 3% median capacity: about 15.6m TWD.
- ADV 5% median capacity: about 26.1m TWD.

Liquidity $\geq$ 100m S1

- ADV 1% median capacity: about 7.6m TWD.
- ADV 3% median capacity: about 22.8m TWD.
- ADV 5% median capacity: about 38.0m TWD.

Interpretation: liquidity $\geq$ 100m helps capacity, but does not automatically improve alpha quality.

05 — Execution Realism and Paper-Trading Plan

Why execution realism matters

Monthly revenue strategies are event-driven. A backtest can look attractive if it assumes entry at a price that was not actually tradable after information became public. The key question is not just "does the signal work?" but:

Could a real researcher observe the data, generate the signal, submit the order, and get a plausible fill without look-ahead or unrealistic queue assumptions?

Current execution-realism work completed

OHLC / limit audit

- Raw official daily JSON files were parsed into a normalized OHLC / limit table.
- TPEx raw files include next-day limit-up/down fields.
- Current local TWSE raw sample has OHLC but lacks parsed next-day limit fields.

Output table:

```
/Users/liuynzhen/quant-research/tw_monthly_revenue/data/processed/official_daily_ohlc_limit_from_raw.csv
```

Conservative entry timing

The project tested:

- entry close proxy;
- next close;
- next open;
- delay 0/1/2/3 trading days;
- possible limit-up non-fill exclusion;
- 0.7%, 1.0%, and 1.5% cost assumptions.

Key conservative result

`boost_quiet_no_large_black_150 | next_open | 1.0% cost | exclude limit-up risk:`

- Return: 111.9%.
- Sharpe: 1.24.
- MDD: -24.9%.

This is still positive, but much less impressive than the entry-close proxy.

Exact-timing gap

The current historical revenue table has:

announcement_date_quality = monthly_summary_no_company_timestamp

This means:

- no exact company-level announcement timestamp;
- no reliable after-close / intraday classification;
- no exact earliest-tradable-date proof;
- no final look-ahead-bias closure.

Paper-trading execution log design

Each future signal should log:

- signal_id
- stock_id
- stock_name
- revenue_month
- data_available_at
- announcement_timestamp_source
- signal_generated_at
- planned_entry_date
- planned_entry_type — open / close / VWAP proxy
- observed_open
- observed_high
- observed_low
- observed_close
- limit_up_price
- opened_at_limit_up
- non_fill_reason
- planned_notional
- ADV_participation_pct
- estimated_slippage
- actual_paper_fill_price
- 5D_return
- 10D_return
- 20D_return
- deviation_from_backtest_assumption

Paper-trading acceptance criteria

Before this project should be described as production-adjacent, it should accumulate multiple monthly revenue cycles with:

- timestamped signal generation;
- documented feasible entries;
- explicit non-fill cases;
- realized paper PnL under predefined assumptions;
- comparison against the historical proxy model;
- explanation for any assumption drift.

Recommended paper-trading rule

Until exact timestamps are available:

Use conservative next-trading-day-after-observed-data entry. Do not claim same-day tradability.

01 — Methodology Report

Research question

Can Taiwan monthly revenue disclosures create a tradable post-announcement drift signal, especially when revenue surprise is persistent and the stock is not already overextended?

Market-structure motivation

Taiwan companies disclose monthly revenue, giving investors a higher-frequency fundamental signal than quarterly earnings alone. The hypothesis is not simply "high revenue growth is good." The more precise hypothesis is:

Because monthly revenue surprise updates investors' expectations about near-term demand and supply-chain strength, and because some investors react with delay after official disclosure, stocks with persistent positive revenue surprise may continue repricing over the next 10 – 20 trading days.

This should be strongest when:

- surprise is persistent, not just one noisy month;
- price is not already overextended;
- the sector has a strong narrative / capital-flow channel, especially electronics and semiconductors;
- liquidity is sufficient for research-level tradability;
- the first feasible trade is after public data availability.

Data used

The project uses local official / official-derived Taiwan data artifacts:

- Monthly revenue panel with `usable_date_proxy` and revenue fields.
- Daily market history with close and turnover value.
- Raw official daily JSON files, later parsed into normalized OHLC / limit research tables.

Important data caveat:

```
announcement_date_quality = monthly_summary_no_company_timestamp
```

This means historical company-level exact announcement timestamps are not yet available in the current dataset. The project therefore uses conservative proxy timing and explicitly refuses to claim production-ready tradability.

Factor construction

The core factor family includes:

- Revenue YoY / MoM.
- 3-month revenue growth / persistence.
- Standardized unexpected revenue / SUR-style surprise.

- Industry-adjusted surprise variants.
- Momentum and overextension filters.
- Abnormal turnover and daily OHLC/K-line diagnostics.

The final incumbent emphasizes:

3M SUR persistence + not-overheated momentum

Portfolio construction

The main S1 fixed-20 comparator uses:

liquidity $\geq$ 50m TWD avg turnover
 Top 8 names per month
 industry cap = 3
 holding period = 20 trading days
 round-trip cost proxy = 0.7% in baseline tests

The portfolio-grade v0.1 incumbent from the broader search adds a trailing/stop-style proxy:

sur3_high_no_high_mom | liq50m | top8 | industry_cap=3 | semi_cap=none | sl8_trail12 | 20D

Research gates

The project deliberately uses promotion gates rather than simply chasing Sharpe:

1. Baseline factor must be coherent.
2. New signals must beat the incumbent under comparable assumptions.
3. Inactive months are counted as cash, not removed from Sharpe.
4. Remove-top-winners tests must be reported.
5. Year split and walk-forward OOS must be checked.
6. Electronics / semiconductor dependence must be tested.
7. Liquidity and cost sensitivity must be tested.
8. Execution realism must be audited before any production claim.
9. Paper-trading schema must exist before live consideration.

Main research evolution

The project evolved through these layers:

- Phase 2: data feasibility and proxy backtest.
- Phase 3.1 – 3.5: portfolio NAV, robustness, multifactor, SUR tests.
- Phase 3.6 – 3.11: short-horizon SUR, execution rules, high-Sharpe search, signal quality search.
- Phase 3.12: walk-forward and sector survival.
- Phase 3.13 – 3.18: price/volume, K-line, quiet digestion, dynamic sizing.
- Phase 3.19 – 3.26: execution realism, delay sensitivity, OOS sector stress, paper-trading schema, normalized OHLC/limit parser.

Final research interpretation

The strongest defensible interpretation is:

A Taiwan electronics / semiconductor supply-chain monthly revenue surprise strategy with evidence of post-disclosure drift, but with meaningful right-tail, sector, data-timing, and execution constraints.

The project is strongest as a research process demonstration, not as a live trading claim.

03 — Research Timeline

Phase 2 — Data feasibility and proxy backtest

- Identified official / free Taiwan revenue and market data sources.
- Built historical monthly revenue panel and initial proxy backtest.
- Established that revenue timing must be handled conservatively.

Phase 3.1 – 3.5 — Portfolio NAV and SUR factor research

- Moved from trade averages to cohort NAV diagnostics.
- Added robustness / risk tests and multifactor extensions.
- Tested SUR-style definitions and industry-adjusted variants.
- Learned that persistent 3M SUR is more promising than raw single-month growth alone.

Phase 3.6 – 3.8 — Short-horizon and execution-rule exploration

- Focused on 10 – 20 trading-day horizons.
- Found 20D fixed horizon more defensible than over-aggressive 10D variants.
- Tested stop/trailing proxies, but labeled them close-price proxies rather than executable stops.

Phase 3.9 – 3.11 — High-Sharpe search and signal-quality discipline

- Broad search did not find a robust Sharpe > 2.5 candidate that survived all gates.
- Preserved S1 as incumbent rather than overwriting it with fragile high-return variants.
- Established promising strategy registry.

Phase 3.12 — Walk-forward and sector survival

- Walk-forward parameter selection did not beat fixed S1.
- Sector survival showed strong electronics / semiconductor dependence.
- Reframed strategy away from broad-market alpha and toward supply-chain revenue surprise.

Phase 3.13 – 3.16 — Price/volume and K-line diagnostics

- Tested abnormal turnover, low-volume digestion, K-line supply pressure, and large black K diagnostics.
- Found volume expansion alone is not a free lunch.
- Found quiet digestion is interesting but sparse and winner-dependent.

Phase 3.17 – 3.18 — Quiet digestion and dynamic sizing

- Deep-dived quiet digestion as low abnormal turnover plus narrow entry-day range.

- Concluded it is not a standalone replacement strategy.
- Tested it as S1 dynamic sizing.
- Best sizing rule modestly improved fixed-20 proxy but did not reduce MDD enough for promotion.

Phase 3.19 – 3.24 — Execution realism and OOS stress

- Added next-open / next-close timing, cost stress, limit-up non-fill proxy, and ADV capacity proxy.
- Walk-forward dynamic sizing produced small OOS improvements.
- OOS sector stress showed improvements are still semiconductor / winner dependent.

Phase 3.25 – 3.26 — Exact timing and paper-trading preparation

- Audited data and confirmed historical revenue table lacks exact company-level timestamps.
- Created paper-trading execution schema.
- Parsed raw official daily JSON into normalized OHLC / limit research table.

Final status

The project is ready to package as a serious quant research portfolio piece. It is not ready to be marketed as a production trading system.

06 — Limitations and Next Steps

Current limitations

1. Exact announcement timestamp is missing

The historical revenue data currently uses a monthly summary / usable-date proxy, not company-level exact announcement timestamps. This is the biggest remaining data-timing limitation.

2. Survivorship and universe completeness need stronger controls

A production-grade version should explicitly handle:

- delisted names;
- suspended names;
- full-delivery / special treatment stocks;
- listing changes;
- historical industry classifications;
- non-common instruments.

3. Execution model remains proxy-based

The project now has OHLC and TPEx next-limit fields, but still lacks:

- full order-book depth;
- opening auction queue position;
- partial fills;
- intraday volume distribution;
- true limit-up queue outcomes;
- broker-level implementation assumptions.

4. Sample length is short

The strongest research period is mainly 2023 – 2025. That is enough for a portfolio project but not enough to claim a durable structural anomaly.

5. Sector dependence is material

Results are meaningfully stronger in electronics / semiconductor supply-chain names. The project should be framed as sector-conditional, not broad-market.

6. Winner concentration remains important

Remove-winner tests often reduce Sharpe sharply. This suggests right-tail dependence and regime sensitivity.

What would make it stronger as a portfolio project?

1. Turn the current documents into a polished public README.
2. Add a clean architecture diagram for the data pipeline.
3. Add one reproducible notebook or script that rebuilds the main robustness table.
4. Add a small sample paper-trading log using future 2026 revenue cycles.
5. Add a clear "what I learned / what I rejected" section.

What would make it stronger as a trading strategy?

1. Obtain exact company-level monthly revenue announcement timestamps.
2. Extend historical sample if reliable official data is available.
3. Improve survivorship and historical universe membership handling.
4. Build a stricter execution simulator with limit-up/down, suspensions, and partial fills.
5. Run live paper trading for several cycles.
6. Re-run all promotion gates under those assumptions.

Current recommendation

Stop expanding the signal grid for now. The highest-value next step is not more K-line mining or parameter search. It is:
exact timing + paper trading + operational feasibility.

07 — Interview Q&A

Q1. What is the core idea?

Taiwan companies disclose monthly revenue, which provides a recurring high-frequency fundamental update. The strategy tests whether stocks with persistent positive revenue surprise experience delayed repricing after public disclosure, especially in electronics and semiconductor supply chains.

Q2. Why use 3M SUR instead of raw revenue growth?

Raw growth can already be priced in and can be noisy. 3M SUR persistence tries to capture a more stable surprise component: repeated upside relative to recent expectation / baseline rather than a single impressive YoY print.

Q3. How did you avoid look-ahead bias?

The project separates revenue month from usable date and uses conservative data-availability proxies. Later phases explicitly stress next-open / delayed entries and acknowledge that exact company-level timestamps are still missing, so the project does not claim production-ready timing.

Q4. What was the best-performing strategy?

The preserved portfolio-grade v0.1 candidate is S1:

```
sur3_high_no_high_mom | liq50m | top8 | industry_cap=3 | semi_cap=none | sl8_trail12 | 20D
```

It had a proxy Sharpe around 2.40, return around 167.5%, and MDD around -7.9% in the earlier portfolio-grade proxy setup.

Q5. Why not just promote the highest Sharpe variant?

Because many high-return variants were fragile after remove-winner tests, sector survival checks, or execution realism. The project uses promotion gates rather than full-sample Sharpe ranking.

Q6. What did price/volume and K-line research add?

It showed that volume expansion alone is not a reliable alpha. Quiet digestion — low abnormal turnover plus narrow entry-day range — had an interpretable delayed-repricing story, but as a standalone slice it was sparse and winner-dependent. It was therefore tested as a sizing diagnostic inside S1 rather than promoted as a replacement.

Q7. What is the biggest weakness?

Exact company-level announcement timestamps are missing in the current historical revenue dataset. That prevents a final answer on same-day tradability and exact no-look-ahead timing.

Q8. Is the strategy broad-market?

No. The evidence points to electronics / semiconductor supply-chain dependence. That is not necessarily bad, but it must be stated honestly.

Q9. What happened under conservative execution assumptions?

Under `next_open + 1.0%` cost + exclude possible limit-up risk, quiet boost dropped to return `111.9%`, Sharpe `1.24`, and MDD `-24.9%`. This is still interesting but no longer a headline high-Sharpe result.

Q10. What would you do next?

I would stop expanding the parameter grid and instead:

1. obtain exact announcement timestamps;
2. run real-time paper trading for 2026 revenue cycles;
3. log feasible fills and non-fills;
4. re-run promotion gates with actual timing and tradability;
5. only then consider whether it deserves production research.

Q11. What does this project demonstrate about your research style?

It demonstrates that I can start from a market-structure hypothesis, build a systematic research pipeline, generate and reject variants, test robustness, preserve promising candidates, and avoid overstating results when operational constraints remain unresolved.

Paper-Trading Template

This template is for research / paper-trading only. It is not a live order ticket.

Monthly workflow

1. Data availability check

- Confirm the monthly revenue source has been updated.
- Record the actual observation time.
- Do not assume a company was tradable before the data was visible.

Required fields:

revenue_month:
data_source:
data_observed_at:
data_timezone: Asia/Taipei
announcement_timestamp_quality:
operator:
notes:

2. Signal generation log

For every candidate generated by the model:

signal_id:
generated_at:
stock_id:
stock_name:
industry:
revenue_month:
usable_date_proxy:
sur_3m_score:
momentum_state:
quiet_digestion_flag:
large_black_k_flag:
rank_in_month:
selected_by_rule:
rule_name:
planned_weight:
planned_notional_twd:

3. Planned execution log

signal_id:
planned_entry_date:
planned_entry_session: open / close / vwap_proxy
planned_entry_reason:
reference_price_source:
expected_open:
previous_close:
limit_up_price:
limit_down_price:
opened_at_limit_up:
opened_at_limit_down:
ADV_20d_twd:
planned_ADV_participation_pct:
max_allowed_ADV_participation_pct:

4. Paper fill log

signal_id:
paper_fill_status: filled / partial / not_filled / skipped
paper_fill_price:
paper_fill_time_assumption:
filled_notional_twd:
non_fill_reason:
estimated_slippage_bps:
estimated_commission_tax_bps:
all_in_cost_bps:

5. Holding-period outcome log

signal_id:
entry_date:
exit_rule: fixed_20d / stop_proxy / trail_proxy / manual_research_exit
planned_exit_date:
actual_paper_exit_date:
exit_price:
return_5d:
return_10d:
return_20d:
max_favorable_excursion:
max_adverse_excursion:
stop_or_trail_triggered:
limit_or_suspension_during_holding:

6. Assumption drift review

After each monthly cycle, answer:

1. Did the real signal list match the backtest rule?
2. Were any names impossible to enter because of limit-up / suspension / illiquidity?
3. Was slippage larger than the model assumption?
4. Did actual data timing differ from `usable_date_proxy`?
5. Did electronics / semiconductor names dominate the selected book?
6. Did one or two winners dominate monthly PnL?
7. Should this month be marked as model-valid, execution-impaired, or data-timing-impaired?

CSV header

signal_id,revenue_month,data_observed_at,generated_at,stock_id,stock_name,industry,rule_name,sur_3m_score,momentum_state,quiet_digestion_flag,large_black_k_flag,rank_in_mo

Acceptance threshold before production-adjacent claims

Do not call the strategy production-adjacent until at least several monthly cycles have:

- timestamped signal generation;
- feasible paper fills;
- explicit non-fill accounting;
- stable slippage / cost assumptions;
- documented performance versus historical proxy expectations;
- no hidden same-day look-ahead assumption.

Reproducibility Guide

Scope

This guide documents where the research artifacts live and how to reproduce the major outputs. It does not place trades, connect to a broker, or install packages.

Important paths

Project root:

/Users/liuynzhen/quant-research/tw_monthly_revenue

Portfolio docs:

/Users/liuynzhen/quant-research/tw_monthly_revenue/portfolio_project

Phase reports:

/Users/liuynzhen/quant-research/tw_monthly_revenue/reports

Charts:

/Users/liuynzhen/quant-research/tw_monthly_revenue/reports/charts

Processed data:

/Users/liuynzhen/quant-research/tw_monthly_revenue/data/processed

Scripts:

/Users/liuynzhen/quant-research/tw_monthly_revenue/scripts

Core scripts

Data and initial research

scripts/fetch_historical_monthly_revenue_mops_static.py

scripts/fetch_daily_market_history_2023_present.py

scripts/run_proxy_backtest.py

scripts/portfolio_nav_backtest.py

SUR and strategy research

scripts/sur_factor_tests.py

scripts/short_horizon_sur_research.py

scripts/execution_realism_tests.py

scripts/high_sharpe_search.py

scripts/signal_quality_search.py

scripts/walkforward_sector_survival.py

Price/volume and K-line research

scripts/price_volume_kline_research.py

scripts/price_volume_sector_diagnostics.py

scripts/kline_ohlc_audit_research.py

scripts/price_volume_kline_interaction_research.py

scripts/quiet_digestion_deep_dive.py

scripts/quiet_digestion_dynamic_sizing.py

Execution realism and paper-trading gates

scripts/execution_realism_tradability_gate.py

scripts/dynamic_sizing_walkforward_gate.py

scripts/exact_timing_delay_sensitivity_gate.py

scripts/exact_timing_robustness_gate.py

scripts/delay_walkforward_robust_selection_gate.py

```
scripts/delay_walkforward_oos_sector_stress.py
scripts/exact_timing_paper_trading_schema_gate.py
scripts/build_official_daily_ohlc_limit_from_raw.py
```

Recommended reproduction order

From project root:

```
python3 scripts/sur_factor_tests.py
python3 scripts/signal_quality_search.py
python3 scripts/walkforward_sector_survival.py
python3 scripts/quiet_digestion_dynamic_sizing.py
python3 scripts/execution_realism_tradability_gate.py
python3 scripts/dynamic_sizing_walkforward_gate.py
python3 scripts/exact_timing_delay_sensitivity_gate.py
python3 scripts/delay_walkforward_oos_sector_stress.py
python3 scripts/exact_timing_paper_trading_schema_gate.py
python3 scripts/build_official_daily_ohlc_limit_from_raw.py
```

Key reports to inspect

```
reports/promising_strategy_registry.md
reports/phase3_12_walkforward_sector_survival_report.md
reports/phase3_18_quiet_digestion_dynamic_sizing_report.md
reports/phase3_19_execution_realism_tradability_report.md
reports/phase3_20_dynamic_sizing_walkforward_report.md
reports/phase3_24_delay_walkforward_oos_sector_stress_report.md
reports/phase3_25_exact_timing_paper_trading_schema_report.md
reports/phase3_26_official_ohlc_limit_parser_report.md
```

Key processed outputs

```
data/processed/execution_realism_tradability_summary.csv
data/processed/dynamic_sizing_walkforward_summary.csv
data/processed/delay_walkforward_oos_sector_stress.csv
data/processed/delay_walkforward_oos_remove_winners.csv
data/processed/paper_trading_execution_log_schema.csv
data/processed/official_daily_ohlc_limit_from_raw.csv
```

Verification commands

```
python3 -m py_compile scripts/*.py
```

For a lightweight file-presence check:

```
test -f reports/promising_strategy_registry.md
test -f portfolio_project/README.md
test -f data/processed/official_daily_ohlc_limit_from_raw.csv
```

Reproducibility caveats

- Some reports depend on previously fetched local raw data.
- Current historical revenue timing is proxy-based, not exact timestamp-based.
- Raw TWSE local coverage is more limited than TPEX coverage in the current artifact set.
- Results are research diagnostics, not trading recommendations.